❖ Final RNS (35)

Histology

1- Endothelium of glomeruli :

A) fenestrated B) has diaphragm C) continuous D) regenerated by macula densa

2- Secretory granules present in :

A) juxtaglomerular cells B) macula densa C) Lacis cells D) mesengeal cells

3- Epithelium of prostatic urethra proximal to ejaculatory duct opening is :

A) transitional B) pseudo stratified columner C) stratified columner D) stratified squamous

Microbiology

4- Recurrent uncomplicated UTI :

A) 3 times or more per year B) 2 times or more per year

C) one time per one year D) another pathogen affect 3 times a month

5- Which of the following urinalysis strongly indicates bacterial infection :

A) positive nitrites B) positive keton

C) low specefic gravity D) presence of RBCs only

Physiology

6- About Angiotensin || :

A) increase renal blood flow B) contracts afferent arterioles more than efferent arterioles

C) contraction of mesengeal cells D) less active than angiotensin |||

7- Low ADH levels :

A) due to supraoptic nuclei lesion B) Lead to formation hypo osmotic urine

C) occurred at end of fasting day D) Lead to increase Na concentration in plasma

8- Micturition reflex when urine pass in urethra produce more contraction of detrusor muscle occurred by :

A) afferent pudendal , center LHC pelvic efferent

B) afferent pudendal , center AHC pudendal efferent

C) afferent pelvic , center AHC pudendal efferent

D) afferent pelvic , center LHC pelvic efferent

9- The reabsorption of filtered HCO3 :

A) results in reabsorption of less than 50% of the filtered load in PCT

B) is directly linked to excretion of H+ as NH4+

C) inhibited by decrease PCo2 in plasma D) acidifies tubular fluid to a pH of 4.4

10- Reduction glucose reabsorption and appearance glucose in urine occurred by process occurs in proximal tubules :

A) saturation Na-glucose co transport B) stimulation of glycogen metabolism

11- Na reabsorption on basolateral membrane of tubules occurred by:

A) Na / K pump B) Na / glucose co transport C) facilitated diffusion D) Na / H exchange

12- Bowman's capsule hydrostatic pressure :

A) increased by pelvis and ureter obstruction B) delivering filtration force

C) more than plasma colloid pressure

13- Medullary hyperosmolarity causes:

A) increased Na+ reabsorption from the DCT B) increased H+ secretion from the PCT

C) increased H2O reabsorption from the collecting ducts D) dilution of urine

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Pathology

14- Common stone occurred on top of infection :

A) oxalate B) uric acid C) cystein D) phosphate

Anatomy

15- What's the Hilum and structures pass through it from anterior to posterior? (2 Marks)

16- Mention congenital anomalies of the kidney (2 Marks)

Pathology

17- Mention pathological features of clear cell RCC (2 Marks)

18- Mention components of nephrotic syndrome (2 Marks)

Pharmacology

19- Give Four therapeutic uses for Torsemide (2 Marks)

20- Mention mechanism of drug induced tubular cell toxicity (1 Mark)

Biochemistry (1 Mark)

21- Alkalinization can be used to different management of ..... and ..... stones

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Parasitology (1 Mark)

22- Plasmodium Malaria causes ..... in children

ANSWERS

1	2	3	4	5	6	7	8	9	10
A	A	A	A	A	C	B	A	C	A

11	12	13	14
A	A	C	D

15-It is a vertical opening at the middle of the medial border at the level of L1 vertebrae.

Structures (VAP) : 1) renal vein 2) renal artery 3) renal pelvis

16-1) Renal agenesis & hypoplasia 2) Ectopic kidney 3) Horse-shoe kidney

4) Congenital polycystic kidney 5) Accessory renal arteries

17- Grossly:

A solid, golden- yellow tumor with variable amounts of cystic change, haemorrhage, calcification and necrosis. The tumor is sharply separated from the surrounding tissue by a fibrous pseudocapsule.

Microscopically:

The cells are large, rounded or polyhedral and arranged in acini or trabeculae separated by scanty stroma containing thin walled blood vessels. The cytoplasm is clear or vacuolated due to high content of lipid and glycogen. The nucleus is small and darkly stained.

18-1) Massive proteinuria 2) Hyperlipidemia 3) Generalized edema 4) Hypoproteinemia

19-1) acute pulmonary edema

2) other edematous states as in Cardiac, renal, or hepatic diseases

3) hypertensive emergency in patients with chronic renal failure

4) hyperkalemia

5) Symptomatic hypercalcemia

20-1) impairing mitochondrial function

2) interfering with tubular transport

3) increasing oxidative stress or forming free radicals

21-uric acid - cystine - calcium oxalate

22-Nephrotic syndrome